

Zakwan Haaziq K

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PROFILE SUMMARY

Computer Science undergraduate who has designed, trained, and deployed 6+ end-to-end Machine Learning and Deep Learning applications across healthcare and computer vision domains using TensorFlow, PyTorch, Scikit-learn, FastAPI, and Streamlit. Skilled across the complete ML pipeline — data preprocessing, feature engineering, transfer learning, hyperparameter tuning, Grad-CAM explainability, and production deployment on Docker, HuggingFace Spaces, and Streamlit Cloud. Experienced building CNN-based classifiers, REST API backends with FastAPI, and real-time inference systems that are live and publicly accessible.

EXPERIENCE

Machine Learning Intern

Aug 2025 – Sep 2025

Novi Tech Research & Development Pvt. Ltd.

- Developed end-to-end **machine learning pipelines** for classification and regression tasks using PyTorch and Scikit-learn, covering **data preprocessing**, **feature engineering**, model training, and evaluation.
- Implemented **Convolutional Neural Networks (CNN)** for computer vision tasks including image classification and object detection, applying data augmentation and **transfer learning** to improve model generalization.
- Optimized models via **hyperparameter tuning** and **k-fold cross-validation**, systematically improving accuracy and reducing overfitting across real-world datasets.
- Deployed trained models as interactive **Streamlit** web applications, enabling **real-time inference** and live user interaction with production ML systems.

Machine Learning Intern

Sep 2025 – Oct 2025

CodSoft

- Implemented 5 ML tasks spanning **binary classification**, multi-class classification, and **regression** using **Scikit-learn**, Pandas, and NumPy on structured real-world datasets.
- Executed end-to-end **data preprocessing** pipelines including missing value imputation, label encoding, **feature scaling**, and outlier removal to ensure clean model inputs.
- Evaluated models using accuracy, **precision**, **recall**, **F1-score**, and ROC-AUC; applied **cross-validation** to verify generalization and prevent data leakage.

EDUCATION

B.E. Computer Science and Engineering

Sep 2024 – Present

C. Abdul Hakeem College of Engineering and Technology, Melvisharam | CGPA: 9.01

Senior Secondary (XII) – Tamil Nadu State Board

Mar 2023 – Mar 2024

K.H. Matriculation Boys Higher Secondary School, Melvisharam | Score: 86.6%

Secondary Education (X) – Tamil Nadu State Board

Mar 2021 – Mar 2022

K.H. Matriculation Boys Higher Secondary School, Melvisharam | Score: 86.8%

PROJECTS

Brain Tumor Detection (NeuroScan AI) | TensorFlow, VGG16, Grad-CAM, FastAPI, Gemini 2.0, Docker | [Live ↗](#)

- Fine-tuned **VGG16** on 7,000+ MRI scans across 4 classes (Glioma, Meningioma, Pituitary, No Tumor) using **two-phase transfer learning** — frozen base for 5 epochs then top-20 layers unfrozen — boosting accuracy from 84% to **96% test accuracy**.
- Implemented **Grad-CAM** on the last convolutional layer (block5_conv3) to generate heatmap overlays that visually highlight tumor regions in MRI scans, enabling clinician trust and **model explainability**.
- Built a **FastAPI** backend with /predict and /caption endpoints; integrated **Gemini 2.0 Flash** via OpenRouter to auto-generate structured diagnostic reports; containerized with **Docker** and deployed on **HuggingFace Spaces**.

MedTrust – Health Consultant Backend | FastAPI, Firebase, Scikit-learn, PyTorch, Docker | [Live ↗](#)

- Engineered a **FastAPI** backend with 4 dedicated ML model endpoints — Heart Disease, Diabetes, Life Risk, and Brain Tumor — with a **model_loader.py** that auto-downloads all models on startup forming a complete **machine learning pipeline**.
- Defined structured request/response schemas for all 4 endpoints using **Pydantic** models (schema/ folder), ensuring robust **input validation** and consistent API contracts.
- Integrated **Firebase** for real-time authentication and built an admin monitoring dashboard for live tracking of user requests; containerized with **Docker** and deployed on **HuggingFace Spaces**.

Heart Disease Prediction Chatbot | Scikit-learn, Pandas, Streamlit | [Live ↗](#)

- Trained and serialized an ML classifier (Model.pkl) on the hearts.csv dataset; applied **data preprocessing** including **feature selection** and handling of clinical variables such as **ST Slope** and **Oldpeak**.

- Implemented smart UX logic that auto-calculates **Maximum Heart Rate** as $(220 - \text{age})$ and internally handles hidden clinical features, reducing user input friction while maintaining prediction accuracy.
- Deployed a conversational chatbot-style **Streamlit** UI that collects patient parameters and serves **real-time** heart disease risk predictions through the serialized ML model.

Face Mask Detection | *PyTorch, OpenCV, Streamlit* | [Live ↗](#)

- Designed and trained a **custom CNN architecture** (defined in architecture.py) using **PyTorch** on a Kaggle face mask dataset; saved best model weights as **best_model.pth** using checkpoint-based training.
- Applied image preprocessing and **data augmentation** using **OpenCV** including resizing to fixed dimensions, normalization, and flipping to improve model robustness across varied real-world lighting and angles.
- Deployed on **Streamlit Cloud** where users upload an image and receive instant **binary classification** — masked vs. unmasked — via **real-time inference** through the loaded PyTorch model.

SKILLS

Programming Languages: Python, Java, C, C++

Machine Learning & Deep Learning: PyTorch, TensorFlow/Keras, Scikit-learn, OpenCV, VGG16, CNN, Grad-CAM

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

Frameworks & Backend: FastAPI, Streamlit, Docker, Pydantic

Tools & Platforms: Git, GitHub, Firebase, HuggingFace Spaces, Power BI, Tableau, MS Excel

ACHIEVEMENTS

- **Smart India Hackathon 2025 (SIH)** — Represented college at national level with a real-time Deep Learning project for wheat disease classification.
- **Academic Excellence** — Maintaining CGPA of 9.01 in B.E. Computer Science Engineering (1st Year).